p3 changelog (lessons 01-24)

p3 part 1: lessons 01-06 (03-react-state-and-hooks)

Sources: react.dev reference pages (useState, useReducer, useEffect, useEffectEvent, createContext, Activity), React 19.2 and 19.3 blog posts, react.dev/versions. Executed: Vitest + RTL + jsdom against react/react-dom 19.3.1, `tsc` 5.9 against @types/react 19.3. Scratch tests: `tools/sandbox/scratch/p3/l01-06/*.test.jsx`, `ts/a.tsx`. Date 2026-10-22.

01 useState internals

- Batched `setN(1)` x3 -> 1 render; repeat click -> bail-out, one extra component call on 2nd click, none on 3rd (matches lesson's hedge): **confirmed**.
- Typing a, b, submit -> 3 renders, final `{draft:'', saves:1}`: **confirmed**.
- Mutate + `setItems(items)` -> no render; appears after unrelated render: **confirmed**.
- Lazy init / eager arg in Strict Mode both called 2x: **confirmed**.
- Setter after unmount logs no warning: **confirmed**. Set-state-during-render: child never sees stale value: **confirmed**.
- Changes: removed the ⚠ on the unmounted-setter note and the "(unverified deep link)" tag (URL returns 200).

02 functional updates

- Fold example 43/42, `votes` walkthrough (40, one render; Strict Mode doubles updater calls `+1,+1,*10,*10` but same state): **confirmed**.
- Rebase example (base 1, transition `+10`, urgent `*2`): rendered values `1, 2, 22`: **confirmed**.
- Function-as-state ambiguity (setter called with a function is an updater): **confirmed**.
- Changes: `useEffectEvent` rows/answers now state "stable since 19.2", Effect-only callable, unstable identity; removed ⚠ legend.

03 useReducer

- Reducer + init double-invoked in Strict Mode: **confirmed** (react.dev + run).
- Reducer used is from the render that processes the queue: **confirmed** (observed), hedge kept.
- **Corrected:** §3.3 claimed an eager reducer evaluation in `dispatch` as for `useState`. In 19.3, three no-op dispatches (reducer returns same state) called the component 3 times (children skipped); there is no eager bail-out for `useReducer`. Rewrote §3.3.
- `<Context value>`: react.dev now calls `.Provider` "legacy"; reworded the version note, removed ⚠.

04 useEffect synchronization

- Chat-room log sequences, prod and Strict Mode (mount/switch/unmount): **confirmed**.
- Discrete event flushes passive effects synchronously (render+effect back-to-back in one microtask flush for a click; later macrotask for timer `setState`): **confirmed**, matches react.dev useEffect page. Removed ⚠.
- Cross-component order (cleanups all first, then setups; child before parent on mount/update): **confirmed**.
- **Corrected:** async effect. Warning text is `useEffect must not return anything besides a function...` plus an "It looks like you wrote useEffect(async () => ...)" hint; and in 19.3 dev the later unmount/re-run throws `TypeError: destroy is not a function` (not a silent no-op). TS error on `Promise<void>`: **confirmed** with tsc.
- Added observed `<Activity hidden>` effect teardown to the Strict Mode section; removed `useEffectEvent ⚠` markers (stable 19.2).
- Deps-array size warning exists ("changed size between renders"): **confirmed** (lesson 05 §3.2).

05 effect dependencies / stale closures

- Ticker prediction (total = 4 at t=4s): **confirmed**. B2 (4 created, 3 cleared while mounted, 4th on unmount): **confirmed**. Stale interval logs the first-render value: **confirmed**.
- **Corrected:** B3 said React throws "Maximum update depth exceeded" for an effect that sets a new array each run. In 19.3 effect-driven loops only emit a dev `console.error` with that text and keep looping (3000 iterations run); the throw ("Too many re-renders") is for render-phase setState.
- `useEffectEvent`: callable from useEffect/useLayoutEffect/useInsertionEffect and other Effect Events, throws if called in render, identity changes every render, stable since 19.2: **confirmed** (react.dev + run). Rewrote the version note and removed all ⚠.
- React Compiler 1.0 released Oct 2025 (react.dev blog list): replaced the "verify" hedge. useMemo cache may be discarded (docs: editing file in dev, suspending on initial mount): **confirmed**.

06 effect cleanup

- Race log order: observed `setup 1, cleanup 1, setup 2, aborted 1, apply 2`; **corrected** the listed order in the lesson (it contradicted its own explanation).
- Probe order `cleanup A, cleanup B, setup A, setup B`: **confirmed**. Passive cleanup sees detached DOM (`isConnected` false), layout cleanup sees it attached: **confirmed**.
- Error in cleanup reaches nearest error boundary: **confirmed**. Anonymous-listener leak counts (3 after 3 cycles; 2 on first Strict mount): **confirmed**.
- Ref callback cleanup (19): returned function replaces `null` call; Strict Mode set, cleanup, set; TS error for implicit-return arrow: **confirmed**.
- **Corrected:** §3.8. `<Activity mode="hidden">` tears down both layout and passive effects and restores them on show (hidden-at-mount runs none). For `<Suspense>` re-suspending visible content only layout effect cleanups ran; passive effects stayed. Lesson had said effects are unmounted in both cases.

- Removed ⚠/verify markers; removed "(unverified deep link)" tags. All URLs in lessons 01-06 return 200 (npmjs.com returns 403 to bots only).

Unresolved

- None of substance. The "eager state" internals in 01 §3.4 remain labelled implementation detail (observed behaviour matches, source not read). Newer-major notes (TS 7, Vitest 5, MSW 3) not applicable in 01-06.

p3 part 2: lessons 07-12 (03-react-state-and-hooks)

Date: 2026-10-22. Sources: react.dev reference pages (useLayoutEffect, useInsertionEffect, useMemo, memo, forwardRef, Profiler, StrictMode, eslint-plugin-react-hooks), React 19 / 19.2 / 19.3 posts, React 19 upgrade guide, React Compiler intro + v1.0 post, MDN (field-sizing, anchor-name). Executed tests: `tools/sandbox/scratch/p3/07-12/{a,b,c,d}.test.tsx`, `ssr.mjs` (React 19.3.1, jsdom, RTL). All 46 URLs in lessons 07-12 were fetched: 45 return 200, npmjs.com returns 403 to bots (unchanged).

07 You might not need an effect

- Derived state via Effect = 2 committed renders per click (`1/0`, `1/2`): confirmed (test a).
- Render-phase `setPrev(n)` unconditional loops to "Too many re-renders" even with unchanged value: confirmed; limit 25 found in react-dom source. Changed "exact limit implementation detail" to cite 25 + verification.
- Design 2 recomputes on unrelated keystroke (1 call), Design 3 `useMemo` 0: confirmed.
- Passive effects flushed synchronously after discrete event: confirmed (runs in microtask right after `click()` returns, before paint). Removed warning marker, added "confirmed".
- `<Activity>`: corrected hedge to stable since 19.2; hidden mode unmounts Effects (19.2 post).
- `useEffectEvent`: marked stable since 19.2 (was "experimental/changing"). Removed all warning markers.
- Unresolved: none.

08 useLayoutEffect

- SSR warning: React <=18 warned; React 19.3 `renderToString` prints nothing and the message string is absent from react-dom-server: corrected from "I believe ... verify" to verified; isomorphic-layout-effect workaround now only for React 18-. Updated the pitfall/E2/Q4 text accordingly. Note: react.dev useLayoutEffect page still carries the old troubleshooting entry.
- Order mutation -> ref -> layout -> passive, cleanup order on unmount (insertion, layout, ref, passive): confirmed (test a).
- Passive flush before paint after discrete event: confirmed, wording tightened (same task, microtask).
- `field-sizing` and CSS anchor positioning: MDN shows Baseline 2026 "newly available" (June 2026 / January 2026). Updated the CSS alternative text and E1 recommendation (anchor

positioning now viable as primary for "latest two versions"; library as fallback).
- Hedged: exact minimum browser versions not checked (told reader to check MDN/caniuse).

09 useInsertionEffect

- Order/refs not attached in insertion effect: confirmed (test a, `ref=null` in insertion).
- **Corrected:** StrictMode's mount cycle does NOT re-run insertion effects (observed: only layout and passive get setup/cleanup/setup). Fixed the order note, B1 answer and the ref-counting remark that claimed "insert -> remove -> insert" under StrictMode.
- Docs caveats added: "may run before or after DOM updated", cleanup/setup interleaved per component.
- "React warns in dev when setting state": dev build contains "useInsertionEffect must not schedule updates." Cited.
- `<style href precedence>` / `<link precedence>` confirmed on react.dev style page.
- Library claim corrected: Emotion uses useInsertionEffect (published package checked); styled-components 6.6.0 does not reference it.
- Removed remaining warning markers (library/framework SSR guidance left hedged).

10 useRef

- `ref` as prop in function components, `forwardRef` still works, render-time ref null on mount: confirmed (test b). forwardRef doc wording: "no longer necessary... will be deprecated in a future release"; text updated.
- Ref callback cleanup, inline callback identity change (cleanup then re-attach; without cleanup: null then node): confirmed.
- StrictMode ref callback extra cycle: confirmed (`node, cleanup, node`; no-cleanup: `node, null, node`), documented on StrictMode page.
- TS implicit-return codemod `no-implicit-ref-callback-return`: confirmed in upgrade guide (was "verify name").
- B1 StrictMode ref-in-render: mount logs `1,2`, click `3,4`: confirmed; replaced "implementation detail, don't rely" with measured values (still labelled impl detail).
- usePrevious returns previous value: confirmed `[0,undefined],[1,0],[2,1]`.
- Compiler refs claim: eslint-plugin-react-hooks docs say compiler skips violating components and has a `refs` rule. Updated; `useEffectEvent` marked stable 19.2.

11 useMemo / useCallback

- StrictMode double-calls calculation (mount and update), 0 calls on unrelated update, React 19 reuses memoized callback from first pass: confirmed (test c); matches useMemo page and upgrade guide.
- Compiler section rewritten from "as I understand ... verify" to docs-backed statements: stable 1.0 Oct 2025; keep existing memoization; useMemo/useCallback remain escape hatches; supports React 17-19. `react.dev/learn/react-compiler` URL verified (200). reactwg/react-compiler URL 200.

- `<Context value>` version note retained, unverified marker removed (verified in 03 owner's lesson 13, not re-run here).

12 React.memo

- memo child receiving a new object prop re-renders (1); with a stable ref callback prop it skips (0); with an inline ref callback it re-renders (1): confirmed (test d). The lesson's claim about React 19 ref-as-prop defeating memo is now stated as verified.
- memo'd context consumer still re-renders on context change: confirmed.
- **Corrected:** Profiler claim: profiling is disabled in the production build by default (react.dev Profiler page); replaced the "verify" hedge.
- Compiler interplay rewritten as in 11.
- `arePropsEqual`/Object.is shallow compare consistent with memo page.

Counts (claims checked)

Confirmed ~34, corrected/updated 11, hedged 2 (browser minimum versions, library SSR specifics), unresolved 0. About 85 warning markers removed across 07-12.

p3 part 3 changelog: lessons 13-18 (03-react-state-and-hooks)

Reviewer date: 2026-10-22. Sandbox: React/ReactDOM 19.3.1, vitest 3.2.7, jsdom, RTL 16. Scratch tests: `tools/sandbox/scratch/p3/l13-18/a13..a18*.test.jsx` (run with `run.sh`). Sources: react.dev reference pages (useContext, createContext, use, useMemo, useEffectEvent, Activity, eslint-plugin-react-hooks, rules-of-hooks), React 19 upgrade guide, React 19.2 / 19.3 / Compiler 1.0 blog posts, plugin README on GitHub.

13 useContext and propagation

- Render log of the theme example (TP, Header, ArticleBody only; not Page/Article/Footer): executed, CONFIRMED. Inline-children variant re-renders Page and Footer: CONFIRMED.
- B1 (`1:default 2:x 3:y 4:x 5:undefined`), B2 (memo consumer re-renders, memo non-consumer does not), B4 (11), conditional `use` after early return, `Context.Provider` still works, Object.is(NaN,NaN)/(0,-0): all CONFIRMED by execution.
- `Context.Provider` deprecation: CORRECTED. react.dev createContext now calls it "a legacy way to provide the context value before React 19"; removed "will be deprecated, verify" hedging in 3 places.
- Legacy `contextTypes`/`getChildContext` removed in 19: CONFIRMED (upgrade guide); removed warning marker.
- NEW finding: a conditional `useContext` does not throw or warn at runtime in 19.3.1 (no hook slot). Added a parenthetical to the useContext-vs-use table (still unsupported/linted).

- Server Components: added 19.3 note (context object can be rendered directly in a Server Component when imported from a 'use client' module). Compiler paragraph updated to 1.0 stable.
- Unresolved: reconciler file names (link kept; marker removed, generic repo path only).

14 Context performance

- 5.1 fat context (3 extra renders after 3 clicks): CONFIRMED. 5.2 split cart (click Add: only CartBadge renders; buttons do not): CONFIRMED. 5.5 selector store (sidebar 0 extra renders): CONFIRMED. Selector returning fresh object: CONFIRMED, React logs "The result of getSnapshot should be cached to avoid an infinite loop" then throws "Maximum update depth exceeded". 5.6 memo reader (Chart 2, Body 1): CONFIRMED. Exercise B1/B2 (A,B / B): CONFIRMED. B3 (batched `add` overwrites; result 2): CONFIRMED.
- CORRECTED: the claim that returning an identical reducer state makes React "bail out entirely, maybe rendering once more": for `useReducer` in 19.3.1 the component function re-ran on each dispatch (children skipped). Reworded.
- Compiler paragraph updated (1.0 stable, 2025-10-07); doc link fixed to /learn/react-compiler/introduction; removed "unverified" markers.

15 Custom hooks

- useDebouncedValue timeline (4 renders over 3 keystrokes + 1 debounced, d effect once): CONFIRMED. useJson in StrictMode makes 2 fetches, first aborted: CONFIRMED. useEventListener latest-ref (handler sees latest, 1 subscription): CONFIRMED. usePersistentState increments via functional set: CONFIRMED. useToggle identity stable via renderHook: CONFIRMED.
- useEffectEvent: CORRECTED from "experimental/verify" to stable since 19.2, with the doc's rules (top-level only, call only from Effects/Effect Events, not passed along, not in deps, identity not stable). Five markers removed.
- ESLint note: `set-state-in-effect` is in the recommended preset (react.dev lint reference): lint note made concrete.
- renderHook availability (RTL >=13.1, present in 16) and anchor: CONFIRMED.

16 Rules of hooks

- Executed in dev build: more hooks ("Rendered more hooks than during the previous render."), fewer hooks ("Rendered fewer hooks than expected. This may be caused by an accidental early return statement."), order-change console error with table, component-called-as-function trap (both directions), hook in event handler ("Invalid hook call"). CONFIRMED.
- CORRECTED/ADDED: effect-before-state removed case gives "Should have a queue. You are likely calling Hooks conditionally..." instead; same-count type swaps: useState/useRef swap renders silently, useState/useMemo swap crashes with TypeError. Table updated.
- CORRECTED F1: lesson claimed `useContext` allocates a positional slot; in 19.3.1 it does not (conditional useContext ran without error). Rewritten to separate the contract (forbidden) from the implementation.
- `use` inside try/catch: docs confirm; dev build also logs a console error, and the catch block receives React's internal Suspense exception for a pending promise. Wording added.

- D2 said the order warning needs Strict Mode: CORRECTED (it does not).
- Effect ordering: all cleanups (children then parent) before any setup in 19.3.1: CONFIRMED, marked version-sensitive.
- ESLint config: added README flat preset (`reactHooks.configs.flat.recommended` , plugin v7.1.1 latest); compiler paragraph updated; link markers resolved (all fetched 200).

17 Derived state

- Effect-synced fullName: 2 renders per keystroke vs 1 derived: CONFIRMED. Mirrored prop stays stale: CONFIRMED. Render-phase setState: body runs twice (3 in StrictMode update): CONFIRMED. Unguarded: "Too many re-renders. React limits the number of renders to prevent an infinite loop.": CONFIRMED. Setting another component's state in render warns "Cannot update a component (`P`) while rendering a different component (`Child`)": CONFIRMED. key reset vs same key: CONFIRMED. useMemo runs twice in StrictMode: CONFIRMED.
- useMemo cache-discard behavior and 1 ms heuristic: CONFIRMED against react.dev useMemo; added the two documented discard cases. Compiler paragraph and link updated.

18 State colocation and lifting

- Scenario 2 (SlowChart 6 renders -> 1 after moving state down), Scenario 4 (children trick, 1 render), temperature converter: CONFIRMED.
- Activity: CORRECTED from "experimental, verify" to stable in 19.2 with documented behavior (display:none, Effects destroyed, state kept, lower priority); sandbox: draft survived, effect setup/cleanup/setup. Interview answer Q5 updated.
- Added concrete lossy round-trip example (213 F -> 101 C -> 214 F). All external links fetched OK; "unverified" note replaced.

Unresolved / not executed

- Internal mechanics (§3 propagation walk, hook list shape): teaching models, hedged in place, not verified against reconciler source.
- Profiler "Context changed" reason labels and DevTools UI steps in §10 of 13/14/18: not executed (no browser).
- Performance figures in prose (4 ms charts, 20 Hz ticks): illustrative, unmeasured.
- Lessons 15/14 later exercise answers (C/D/E/F levels) were read for plausibility, not each executed.

p3 part 4 changelog: lessons 19-24 (react-mastery/03-react-state-and-hooks)

Reviewed 2026-10-22 against react.dev (useState-family reference pages, react-dom hooks, React 19 / 19.2 / 19.3 blog posts) and react/react-dom 19.3.1 in the sandbox. Scratch tests: `tools/sandbox/scratch/p3/p19-24/*.test.tsx` (run with `npx vitest run` from the sandbox dir).

22 Modern hooks and APIs

Claim	How	Result
useEffectEvent stable (19.2), callable from useEffect/useLayoutEffect/useInsertionEffect/other Effect Events, identity unstable, not in deps	react.dev useEffectEvent page + 19.2 post	confirmed; removed hedges
useFormStatus returns `{pending,data,method,action}`, parent-form only, from `react-dom`; method default `get`	react.dev page; sandbox (pending=true, method=get, data=FormData; parent component pending=false)	confirmed
useActionState signature/permalink, queue order, throw cancels queue, not double-invoked in StrictMode	docs; sandbox (3 dispatches run sequentially with prev 0,1,11; single final render)	confirmed
Form reset after Action	React 19 post; sandbox (input returns to defaultValue)	confirmed
useDeferredValue `initialValue`	docs; sandbox (client logs `INIT` then value; renderToString emits only initialValue)	confirmed, wording sharpened
useDeferredValue inside a Transition	docs caveat; sandbox (`v=y d=y`, no extra render)	**corrected/added** (lesson omitted it)
Post-`await` setState not in transition	docs (still present in 19.3); sandbox	confirmed, hedge removed
useOptimistic: "update real state inside Action -> one commit, no flicker"	sandbox, 4 variants (transition fn, form action, wrapped/unwrapped)	**corrected**: unwrapped post-await `setTodos` gives `[] -> [a*] -> [a, a*] -> [a]` (duplicate row for a frame); wrapping in `startTransition` gives `[] -> [a*] -> [a]`. Example code, walkthrough, mermaid, troubleshooting entry and the wrong "item flashes away" explanation fixed
useOptimistic outside Action warns; message text	react-dom dev bundle	confirmed
useActionState outside-transition dev error text	react-dom dev bundle	confirmed

Claim	How	Result
useId prefix	19.2 post + sandbox (`_r_0_` client, `_R_1_` renderToString)	confirmed/clarified (`:r:` 19.0, `«r»` 19.1, `_r_` 19.2+); added doc caveat: don't use useId for `use()` cache keys
use(): no try/catch, promise must be cached, context not supported in Server Components, serializable promise values	docs	confirmed
Uncached `use(new Promise)` in client render	sandbox: suspends (act warnings); not asserted further	partially checked
Typecheck of code blocks	tsc on extracted blocks	no API type errors (syntax-only failures are intentional signature snippets)
Banner replaced with a verified version note (19.3 added ViewTransition and Fragment refs, no changes to these hooks). All warning markers removed.		

20 useSyncExternalStore

- getSnapshot not cached: sandbox confirmed "The result of getSnapshot should be cached to avoid an infinite loop" then "Maximum update depth exceeded" (confirmed).
- Missing getServerSnapshot on server: "Missing getServerSnapshot, which is required for server-rendered content" (confirmed).
- Hydration with different client snapshot: no mismatch error, re-renders to client value (confirmed).
- StrictMode subscribe/unsubscribe balances to 1 active subscription (confirmed).
- Selector store example: coupon change causes 0 re-renders of the badge, qty change 1 (confirmed).
- Sync/non-transition caveat and docs-recommended second getSnapshot call before DOM commit (confirmed against docs; added the second-call detail).
- Zustand 5 and react-redux 9 use useSyncExternalStore (grep in node_modules); sentence rewritten from a hedge to a statement.
- Removed 6 warning markers and a "(verify URL)".

24 Phase assessment

- B1 (3, one extra render), B2 order, B3 (StrictMode `connect a, disconnect a, connect a` then `disconnect a, connect b`), B4 (`timeout 0`, `timeout 1`), B5 (only Row b), B6 (no log for constant value; log for object value), B7 (uncached error then max depth), B8 (3, prev 0 then 1, sequential): executed, all confirmed.
- C5 form-reset behaviour: executed, confirmed (cleared without defaultValue, shows returned value with it); hedge replaced by a verified note.
- useEffectEvent availability note updated (stable since 19.2). Markers removed.
- Unresolved: D/E rubric answers are judgement content, not executed.

21 Forms and complex state

- Uncontrolled reset after Action: documented in React 19 post; sandbox confirmed, including after a returned validation error.
- `requestFormReset(form)`: exists in react-dom 19.3.1, works inside `startTransition` (executed); no dedicated react.dev reference page (the guessed URL 404s), now stated.
- Thrown error in a form Action goes to the Error Boundary (executed).
- `defaultValue={state.email}` pattern works without a `key` (executed); the `key` hack is now described as optional.
- Form-library notes: URLs resolve (React Hook Form, TanStack Form via /form/latest); removed vague "verify" tags. Library behaviour claims unresolved (not tested).

23 Hooks anti-patterns

- React Compiler: 1.0 stable since Oct 2025, opt-in, React 17+ compatible, useMemo/useCallback remain an escape hatch (react.dev blog + intro page). Updated 5 hedged mentions.
- useLayoutEffect on server: renderToString emits no warning in 19.3.1 (executed); text updated.
- Inline ref callback: `el, null, el` across re-render; with a cleanup return the cleanup replaces the null call (executed).
- renderHook is exported by RTL 16 (checked types).
- B4 render-phase setState answers executed: `setA(x)` unconditional throws "Too many re-renders" even when equal; guarded pattern renders twice and settles (confirmed).
- Link `react-developer-tools` resolves (200).

19 UI, server, URL and shared state

- Hydration with differing server snapshot (used by the hand-rolled URL hook): confirmed no mismatch error.
- Navigation API: MDN shows "Baseline 2026, newly available"; text updated.
- Router/library API claims (React Router, Next, TanStack Router) not executed: **unresolved**, left as general pointers to current docs. TanStack Query "section names" tag removed (not checked).